IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

POWER ELECTRONICS AND POWER SYSTEMS

Friday, 8 May 2026, 2:30 PM

Time allowed: 2:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equation for buck SMPS in continuous mode $\frac{V_O}{V_I} = \delta$

Voltage ratio equations for boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{1}{1-\delta}$

Voltage ratio equations for buck-boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{-\delta}{1-\delta}$

Three-Phase Systems (Balanced)

Line/Phase Voltages and Currents

Star $V_L = \sqrt{3}\, V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\, I_P$

Power $P_{3\phi} = 3\, V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Transformers

Ideal transformer $\bar{V}_2 = k_N \bar{V}_1$; $\bar{I}_2{}' = k_N \bar{I}_2$

Impedances referral $\bar{Z}'_2 = {\bar{Z}_2}/{k_N{}^2}$

Voltage regulation $\Delta V_2 = \frac{\left|V_2^{o/c}\right| - |V_2^{load}|}{\left|V_2^{o/c}\right|}$

Induction Machines

$\omega_S = {\omega_E}/{P}$; $\quad s = \frac{\omega_S - \omega_R}{\omega_S}$; $\quad T_{EM} = \frac{3\, I'^2_R\, R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi\, r_C^2} + \frac{1}{\sigma_C 2\pi\, r_O\, t_O}$ $\quad L' = \frac{\mu}{2\pi} ln\left({r_O}/{r_C}\right)$ $\quad C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({r_O}/{r_C}\right)}$ $\quad G' = \frac{2\pi\sigma_I}{ln\left({r_O}/{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2}$ $\quad L' = \frac{\mu}{\pi} ln\left({d}/{r_C}\right)$ $\quad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({d}/{r_C}\right)}$ $\quad G' = \frac{\pi\sigma_I}{ln\left({d}/{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}}\cos(\theta) - \frac{V_R V_S}{Z_{SR}}\cos(\theta+\delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}}\sin(\theta) - \frac{V_R V_S}{Z_{SR}}\sin(\theta+\delta)$$

Power Flow assuming inductance-dominated impedance

$$P_S \simeq \frac{V_R V_S}{X_{SR}}\sin(\delta) \qquad Q_S \simeq \frac{V_S}{X_{SR}}\left(V_S - V_R\cos(\delta)\right)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\, P_S + X\, Q_S}{|V_S|}$

Question 1 – Transformers. This question is worth 25 marks.

a) Some written questions on transformers

i) Explain whether a transformer can operate with a direct current (DC) supply. In your answer, provide a clear justification for your reasoning. [2]

ii) Name a transformer that allows adjustment of voltage levels and explain how this adjustment is carried out. [2]

b) A 50 Hz single-phase step-down transformer has a rated primary voltage of 2.4 kV. The rated primary current is 41.67 A and the rated secondary current is 416.7 A

i) Calculate the turns ratio. [1]

ii) Calculate the rated secondary voltage. [1]

iii) Hence determine the rated apparent power of the transformer. [1]

c) For the transformer in (b), rated voltage is applied to the primary side and the secondary side is open circuit. The open-circuit test results are:

Input voltage	$V_0 = 2.4$ kV
Input current	$I_0 = 0.5$ A
Input power	$P_0 = 724$W

i) Calculate the **approximate** magnetizing loss of the transformer. [2]

ii) Calculate the power factor stating whether it is lagging or leading. [1]

iii) Calculate the magnetizing resistance R_M. [2]

iv) Calculate the magnetizing reactance X_M. [2]

d) For the transformer in (b), rated voltage is applied to the primary side. Under load, the secondary current lags the secondary terminal voltage by $11.3°$, and the secondary current magnitude is 410.0A. The transformer input power is 93.9 kW, and the efficiency is 98.5%.

i) Calculate the output apparent power. [2]

ii) Calculate the real and reactive power absorbed by the load. [3]

iii) Calculate the secondary terminal voltage. [3]

iv) Determine the load impedance connected to the secondary side. [3]

Question 2 – Three-Phase Synchronous Generators and Inverters. This question is worth 20 marks.

a) A three-phase, 600 kVA synchronous generator is star connected to an infinite bus at 3.3kV (line-to-line) and 50Hz. The generator has 2 pole pairs. (Assume balanced steady-state operation. Unless otherwise stated, neglect mechanical losses, iron losses and stray losses. Also neglect armature resistance.)

i) Calculate the synchronous speed of rotation in revolutions per minute. [2]

ii) At synchronous speed, the shaft torque is $3581\ \mathrm{N \cdot m}$. Determine the generator real power output. [2]

iii) The shaft torque is then increased to $3760\ \mathrm{N \cdot m}$ while the machine remains connected to the infinite bus. Determine the new real power output of the generator. [2]

iv) For this increased torque condition in (ii), the power factor is unity and the internal generator emf is 2.1 kV RMS per phase. Calculate the synchronous reactance X_s per phase. [2]

v) The excitation current is increased by 10%. Assume that the internal EMF is directly proportional to the excitation current in this operating region. If the internal EMF before the change is 2.1 kV RMS per phase, and the real power output remains unchanged, determine the magnitude of the reactive current component per phase, and state whether it is supplied or absorbed by the generator. [3]

b) A three-phase permanent magnet synchronous motor (PMSM) is driven by a three-phase inverter.

i) A sinusoidal PWM inverter operates with a switching frequency of 10 kHz and an output fundamental frequency of 50 Hz. Determine the modulation ratio. Is the obtained frequency modulation ratio reasonable? Briefly justify your answer. [2]

ii) The inverter has dc input voltage (rail-to-rail) of 1200 V, which is equivalent to $V_{DC+}=V_{DC-}=600V$. Write the three sinusoidal PWM modulator functions required to generate a 50 Hz three-phase inverter voltage with a phase fundamental peak voltage of 300 V. [2]

iii) At 1500 RPM, the motor back-EMF is 180 V line-to-line RMS. Assume the stator impedance voltage drop is negligible, so that the inverter fundamental output line-to-line RMS voltage is equal to the motor back-EMF. If the dc input voltage is 1200 V and the modulation index is 0.90, determine the motor speed at which this modulation index is required. [2]

iv) Based on iii), add a third-order harmonic component with an amplitude equal to ⅙ of the fundamental modulating signal, as illustrated in Figure 1. Recalculate the modulation index and explain the benefit of third-harmonic injection. [3]

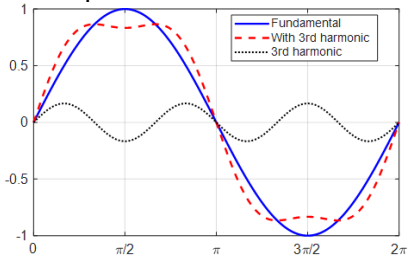


Figure 1. Fundamental modulating signal and its third harmonic-injected waveform

Question 3 – Power Flow in Lines and Cables. This question is worth 25 marks.

a) Consider the characteristics of single-phase overhead lines.

i) An overhead line operates at 50 Hz. Assume the line is lossless and is represented only by its series inductive reactance (resistance and shunt capacitance neglected). The magnitudes of the sending-end and receiving-end voltages are both maintained at 400 kV. The line inductance is $100\ \mu\text{H/km}$. If the maximum transmittable real power is 20 GW, determine the line length. [3]

ii) For the line in i), under the same assumptions, if the allowable maximum current is 16.2 kA, determine the load angle δ corresponding to this current limit. [3]

iii) Series compensation is an effective way to increase the loadability of a transmission line. By inserting a capacitor in series with an inductive transmission line, the total reactance can be reduced. Based on the line parameter in i), calculate the value of the series capacitor that reduces the total reactance of the transmission system (line + capacitor) to half of the original line reactance. [3]

iv) Based on iii), calculate the maximum transmittable real power if the series capacitor is added, and explain the impact of series compensation in increasing the loadability of the line. [4]

v) An overhead line is supplied at 15 kV. The voltage at the receiving end is 14.3 kV. The real power flowing into the line at the sending end is 11.0 MW, and the sending-end power factor is 0.90 leading. The sending-end voltage phasor leads the receiving-end voltage by 20°. Estimate the series impedance of the line. [4]

b) Consider the characteristics of single-phase overhead lines suppling an impedance load, as shown in the figure below, and answer these questions.

i) The overhead line has an impedance of 0.5 + j4.0 Ω. The voltage at the sending end is 1300 V. A pure resistive load R_{L} is connected at the receiving end. If the magnitude of the receiving-end terminal voltage is 1218V, determine the value of R_{L}. [3]

ii) For the line in i), consider a pure resistive load R_{L} connected at the receiving end. Calculate the maximum power that can be supplied to the load. (Hint: find the power delivered to R_{L} as a function of R_{L} and identify the maximum value of the function.) [5]

Question 4 – Switch-Mode Power Supplies. This question is worth 30 marks.

a) A Buck switch-mode power supply (SMPS) is to be designed for an input voltage of 36 V and an output voltage of 12 V. The converter must deliver a maximum output power of 48 W, and the peak-to-peak output voltage ripple must not exceed 60 mV.

Ignore the MOSFET conduction losses. The turn-on and turn-off energy losses of the MOSFET are 2.5 μJ and 3.5 μJ respectively when operated at 1 A, and scale linearly with current. The converter initially operates at a switching frequency of 40 kHz.

i) Design the inductance such that, at maximum output power, the peak-to-peak inductor current ripple ΔI_{L} does not exceed 20% of the average output current. [3]

ii) Assume a film capacitor is used at the output, with sufficiently small ESR such that the ESR-induced ripple can be neglected. Design the minimum output capacitance such that the peak-to-peak output voltage ripple ΔV_{o} does not exceed 60 mV. [3]

iii) If an electrolytic output capacitor is used, neglect capacitive ripple and determine the maximum ESR allowed such that for the peak-to-peak output voltage ripple ΔV_{o} does not exceed 60 mV. [3]

iv) At maximum output power, verify whether the initial switching frequency of 40 kHz satisfies the design requirement that the MOSFET switching loss must not exceed 1.0 W. [3]

v) Then consider reducing the switching frequency to 20 kHz. Recalculate:

- the MOSFET switching loss,
- the inductance required to keep the same ΔI_{L} as in i),
- the capacitance required to keep the same ΔV_{o} as in ii).

Comment on the design implications. [4]

vi) For a purely resistive load, derive the transfer function from the switch-node voltage v_{sw} to the output voltage v_o of the output RLC network and explain, in the frequency domain, how it behaves as a low-pass filter that attenuates switching ripple. [4]

b) A second SMPS of Buck-Boost form is required to produce a -20 V output from the 120 V input. The converter operates at a switching frequency of 50 kHz in discontinuous conduction mode (DCM).

At an output power of 200 W, the duty cycle is 0.2.

The MOSFET to be used has the following switching characteristics:

Turn-on energy loss E_{on} = 12.0 μJ

Turn-off energy loss E_{off} = 18.0 μJ

The MOSFET junction temperature must not exceed 70°C. The ambient temperature is 40°C, and the effective junction-to-air thermal resistance is

$$R_{\mathrm{th}} = 15 \text{ K/W}$$

i) Calculate the energy that must be stored in the inductor at the end of the on-time in each switching cycle in order to transfer 200 W output power. [2]

ii) Calculate the required inductance, and the inductor current at the end of the on-time. [4]

iii) Using the operating condition above, determine the maximum allowable MOSFET on-state resistance $R_{\mathrm{DS(on)}}$ so that the thermal design requirement is satisfied. [4]